Subject: Fwd: Guidelines for MSc thesis writing
Date: Tuesday, 27 August 2024 at 17:58:54 Central European Summer Time
From: Aryanaz Attarpour
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Fyi

Sent from Outlook for iOS

From: Mëmëdhe Ibrahimi <memedhe.ibrahimi@polimi.it>
Sent: Monday, August 5, 2024 11:25:26 AM
To: Bita Rahmat Zadeh <bita.rahmat@mail.polimi.it>
Cc: Massimo Tornatore <massimo.tornatore@polimi.it>; Aryanaz Attarpour <aryanaz.attarpour@polimi.it>
Subject: FW: Guidelines for MSc thesis writing

Dear Bita,

Please make sure to carefully check the thesis instructions that I have provided below.
The main contact during the thesis writing will be me and Aryanaz.

BR,

Mëmëdhe Ibrahimi, PhD
Assistant Professor, Politecnico di Milano
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From: Mëmëdhe Ibrahimi <memedhe.ibrahimi@polimi.it>
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Subject: Guidelines for MSc thesis writing

Dear students,

Please find below some guidelines on how to organize and write your thesis. **Make sure to read it entirely and thoroughly!**

You **should write your thesis in Overleaf**, and you can find the templates in the following link: https://www.ingindinf.polimi.it/en/students/lessons-and-exams/bachelors-and-masters-degree-exams/thesis-format-templates

Depending on the novelty of the work, there are two formats for writing a thesis.

- If you are doing a full thesis, in which your research work goes beyond the state-of-the-art, then you need to submit: 1) the complete thesis, and 2) an executive summary.
- If you are doing a thesina, in which the contribution does not go beyond the state-of-the-art, you just need to submit your thesis

Some good practices (golden rules) when writing the thesis (order is random).

- After each chapter that you write, please go back to this document to ensure you are following the guidelines.
- You need to be clear in what you are saying. Please read and re-read what you write.
- When using figures, make sure to refer to the figures properly. Make sure to comment the figure properly (e.g., avoid sentences like "Fig. 1 shows an optical network"). You need to describe every component that you are showing in the figure, ideally also providing a legend for the figure.
- Use the proper number of references (e.g., no less than 30, and ideally more).
- Please make sure the font in the figure is legible and large enough. Please make sure the size of the figure is large enough.
- Please make sure that the caption of the figure is descriptive. Ideally, one should be able to understand the figure, without having to go and check the description in the text.
- Make sure to use sub-sections, bullet-points, and lists to make the reading easier.
- When you are addressing the comments from a professor in Overleaf, in case you have addressed the comment, reply in the overleaf comment section by saying that you have addressed it. Otherwise, if you do not understand the question/suggestion, ask. Do not remove comments from the professors in overleaf.
- If you resolve an issue in a part of the document, make sure that in case the same issue needs to be resolved in other chapters, you do so accordingly.
- Make sure you upload in the overleaf also the original documents that you use for figures, e.g., if you use PowerPoint. This way, those who review the thesis can have access to modify the figures you provide.
- Make sure to justify and clarify the assumptions you have made when conducting the numerical evaluations. The choice of topology, traffic matrix, and metrics needs to be well-thought and justified. In case of doubts, reach out to your reference contact (PhD student and/or professor following your thesis development).
- **Make sure that you read and re-read what you write!!!**

The organization of your thesis might vary depending on type of thesis you develop, and the topic that you work on. However, there are some general guidelines (good practices) that apply to all thesis. In the following, I will list how most MSc thesis writing is organized.

- Chapter 1: Introduction
  - In this chapter, you are expected to provide a motivation on why is the work that you are doing important, why is it important to solve the problem you are solving.
  - In this chapter, you need to explicitly clarify (maybe through a sub-section) what is the contribution of your thesis.
  - You should clarify in a sub-section, how is the thesis organized by listing the chapters of the thesis, and provide a quick description for each chapter.
- Chapter 2: State-of-the-Art
  - In this chapter, you are expected to overview the state of the art (related works that have addressed the same/similar problems to the one you are addressing).

- You are expected to make it clear how is your work different from the prior works that you are citing and referring to.
- You may also use this chapter to overview some background, depending on the topic you are working on. Please check with your reference contact (with the professor following you, to agree if this part on the background goes in Chapter 2 or Chapter 3).

Chapter 3: Problem statement and problem description

- In this chapter, you are expected to formally write and describe the problem that you are solving. You are expected to clearly highlight the inputs of the problem, the outputs of the problem, the constraints of the problem, and say what is the objective function.
- In this chapter, you are expected to describe the problem you are solving, providing a description of the system model, or the architecture you are considering, or the dataset you are using, ...(there may be other, depending on the topic)....

Chapter 4: Methodologies

- In this chapter, you are expected to describe the methodologies you have developed to solve the problem you are addressing.
- If the methodology you have developed is well-known, e.g., you are using a well-known optimization algorithm such as a Genetic Algorithm, then you are expected to provide a general description, and then say how does it apply (how did you adjust it) to be used for the problem you are solving. As most methodologies are something known from before, you should find a way to provide a general description of the methodology, and then describe it how you developed for your problem.
- It is expected that you provide flowcharts, pseudocode, in this chapter, and provide proper description of such flowcharts/pseudocode.
- You are expected to also describe the baseline (benchmark approach) that you have used as a comparative approach to your proposed solution

Chapter 5: Numerical results

- In this chapter, you are expected to describe what are the methodologies you are comparing (by referring to Chapter 4), in terms of what metrics are you comparing the methodologies, in which topologies/scenarios you are comparing the considered methodologies.
- You are expected to provide numerical results, in terms of plots/tables, and discuss the results that you are reporting properly, providing both a quantitative but also qualitative discussion of the results you are showing. You are also expected to say what are the main takeaways.
- Make sure to organize the chapter is several sub-sections in case you are showing results for multiple scenarios/topologies/other....

Chapter 6: Conclusion and future work

- In this chapter, you are expected to clearly state what are the main takeaways and outcomes of the work you have done in your thesis.
- You need to also list some possible future directions, in which you could extend your thesis

Other thesis sections

- Make sure to have a good number of references, and do not plan to do this in the last moment.
- You need to have an abstract in English and in Italian (at the beginning of the thesis)
- You need to have a list of Figures and list of Tables

Suggestions on writing: It is a good practice that you can start with the writing of Chapter 2 and Chapter 3 first. Then you move to Chapter 4 and Chapter 5. Then you write Chapter 1 (introduction), and then finalize with Chapter 6 (Conclusion).

The same guidelines can be found here MSc thesis guidelines.docx
Good luck and enjoy your vacations!
-Mem

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